MSG No. 72.543 $Ib'a'm$ [Type III, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[11,12,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[13,14,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[0,\ \frac{1}{2},\ 0\right]$	`[13,14,15,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,12]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,10]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[7,13]`

8	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	[8,14]

Table 6: Wyckoff site: 8f, site symmetry: 2’..

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,5]
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[2,6]
3	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[3,7]
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	[4,8]
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[9,13]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[10,14]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[11,15]
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[12,16]

Table 7: Wyckoff site: 8g, site symmetry: .2’.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	[1,6]
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	[2,5]
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	[3,8]
4	$\left[0,\ y,\ \frac{3}{4}\right]$	[4,7]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[9,14]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[10,13]
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[11,16]
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[12,15]

Table 8: Wyckoff site: 8h, site symmetry: ..2

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2]
2	$[0,\ 0,\ -z]$	[3,4]
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[5,6]
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[7,8]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,10]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,12]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[15,16]

Table 9: Wyckoff site: 8i, site symmetry: ..2

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,2]
2	$\left[0,\ \frac{1}{2},\ -z\right]$	[3,4]
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[5,6]

continued ...

Table 9

No.	position	mapping
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ 0,\ z\right]$	`[15,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,4]`
2	$[-x,\ -y,\ 0]$	`[2,3]`
3	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[5,8]`
4	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[6,7]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,11]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[13,16]`
8	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	`[14,15]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-x,\ -y,\ -z]$	`[3]`
4	$[x,\ y,\ -z]$	`[4]`
5	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[6]`
7	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[12]`
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	`[14]`
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[15]`
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[16]`